Sequential Testing for A/B Experiments

A Focused Guide in Three Acts

The problem. Standard hypothesis tests guarantee a controlled false positive rate only when results are analysed once, at a pre-specified sample size. In practice, teams routinely monitor experiments and peek at interim results — inflating the Type I error rate well beyond the nominal α.

The solution. *Sequential testing* provides statistical methods whose error guarantees hold regardless of the number or timing of analyses. The confidence intervals become wider, but it can still reduce overall runtime if effects become clear at an early stage.

This guide. Three acts, each with an interactive simulation, technical exposition, and key formulas:

1. **The Peeking Problem** — Why interim analyses invalidate standard tests.
2. **The Eppo Solution**— Confidence sequences, variance reduction, and the recommended **hybrid approach**.
3. **DIY Alternatives**— Bonferroni, Pocock, and O'Brien–Fleming corrections for teams without a sequential testing platform.

Target audience. Data scientists and analysts familiar with A/B testing, hypothesis testing, and confidence intervals.

Act 1 — The Peeking Problem

Peeking without adjustments

Simulation

Simulate an A/B test. Specify the effect size (difference in means between treatment and control), the number of users in the test, and the significance level (alpha). Optionally, set the desired power (1 - beta). The plot shows the running difference in means and the standard 95% CI. The box below reports, under the null (effect = 0), how often the standard CI crosses the “reject” threshold at some point during peeking.

Effect: (B − A) / A, null hypothesis = 0%.

Effect size: +0.00

Number of users (n = 500)

α (0.05)

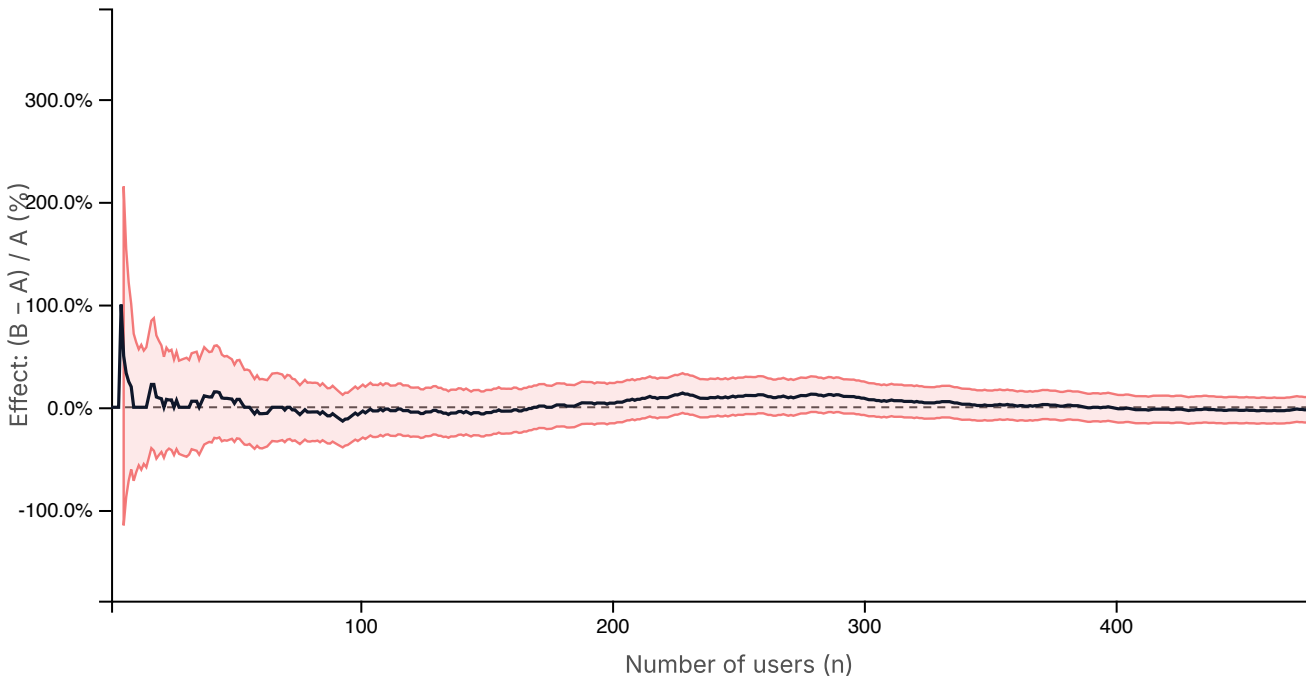
Effect size (+0.00)

Power (1 - β) (0.80)

Re-randomize

Standard 95% CI

Sample diff



Simulation takeaway. With no true effect (effect = 0), the standard CI is calibrated to fail to cover only $\alpha = 5\%$ of the time at one specific look. But if you peek along the way, far more than 5% of trajectories will cross the boundary at some point. The box below shows the share of 500 runs in which the estimate was significant at least once during the test (false positive, leading to early stopping with the wrong conclusion). If you set the effect away from zero, the box below reports the probability of crossing the CI at some point under the simulated effect.

Probability of crossing the CI at some point: `(see simulation output above)`

Why Does This Happen?

Standard confidence intervals only control the chance of a false positive if you decide your sample size in advance and look once at the end. If you check your results along the way, that guarantee no longer holds. Each time we check, we give ourselves another chance to get lucky. The tests are not independent, but the more often you peek—and especially if you stop as soon as you see a “significant” result—the higher your actual false positive rate becomes.

How Bad Is It?

Checking schedule	Actual false positive rate
Once at the end (as designed)	$\sim 5\%$
Daily for 1 week	$\sim 13\%$
Daily for 2 weeks	$\sim 19\%$
Daily for 4 weeks (28 looks)	$\sim 25\%$
Continuously (every observation)	Can exceed 30%

With daily checks over a 4-week experiment, about one in four “significant” results

Key Takeaway

Sequential methods control the probability of ever making a false positive, no matter when we stop. They use stricter or time-dependent thresholds to account for

repeated checking. This way, we can monitor results as often as we like, and the error rate stays controlled.

Act 2 — The Eppo Solution

(Schmit & Miller, 2024)

Simulation

Simulate an A/B test with no true effect (effect = 0). The plot below adds Eppo's **sequential confidence interval**(blue band) on top of the standard 95% CI (red band) you saw in Act 1. The sequential CI is wider at any single look — that is the price of peeking — but its 95% coverage holds *simultaneously at every look*.

Effect: (B − A) / A, null hypothesis = 0%.

Effect size: +0.00

Number of users (n = 500)

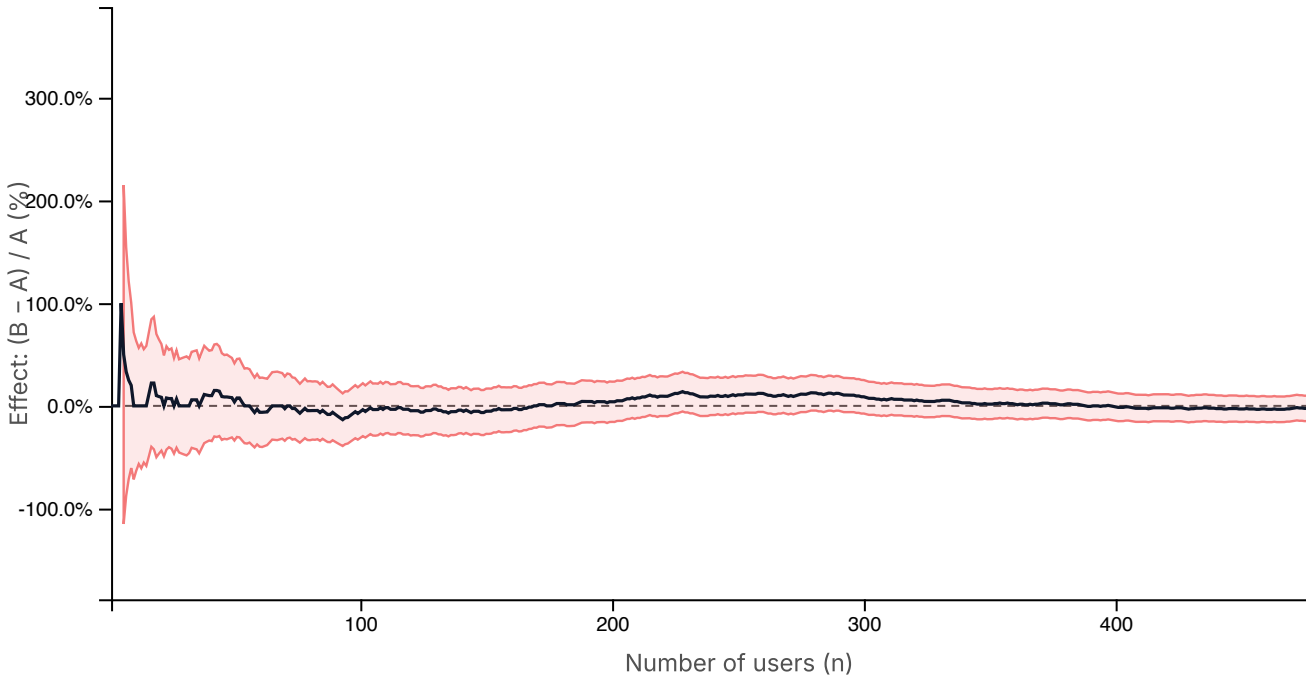
α (0.05)

Effect size (+0.00)

Re-randomize

Standard 95% CI

Sample diff



SAMPLE EFFECT	CI HALF-WIDTH	DECISION AT N = 500
−3.50%	**±12.06%**	Inconclusive

Simulation takeaway. With no true effect, the standard CI (red) crosses the null line in many trajectories — the peeking problem. The sequential CI (blue) is wider but its boundary is calibrated to keep the false positive rate at α across *every* peek. If you set the effect away from zero, the stat box below reports the probability of crossing the CI at some point under the simulated effect.

The Pipeline at a Glance

🎲 Randomise · 📊 Collect · ⚖️ Adjust · 📈 Estimate · 📏 CI · ✅ Decide

The Pipeline, Step by Step

Step 1: Randomise

Users are randomly assigned to **control** or **treatment**. Randomisation ensures exchangeability between groups so that any observed difference in outcomes can be attributed to the treatment.

Step 2: Collect data over time

As users interact with the product, two quantities are recorded for each user:

- Y_i: the **outcome** during the experiment (e.g. purchase amount, clicks, session length).
- X_i: a **pre-experiment covariate** — the same metric from before the experiment started (e.g. purchases in the prior 2 weeks).

Step 3: Regression adjustment (variance reduction)

Individual-level outcome variance is typically large relative to the treatment effect. Eppo reduces this variance by regressing out the predictable component using pre-experiment covariates.

For each group **separately**, fit a model predicting Y from X:

$$\hat{f}_{\text{control}}(X), \qquad \hat{f}_{\text{treatment}}(X)$$

Then compute the **adjusted outcome** for each user:

$$Y_i^* = Y_i - \hat{f}_g(X_i)$$

where $g$is the user's group (control or treatment).

> "Predict what each user would have done based on their history, then subtract that prediction. What's left is the part you *couldn't* have predicted — and that's where the treatment effect hides."
>
> This is a generalisation of a technique called CUPED (Deng et al., 2013). The stronger the correlation ρ between pre-experiment and experiment behaviour, the more noise is removed:
>
> $$\text{Var}(Y^*) = \text{Var}(Y)(1 - \rho^2)$$
>
> At $\rho = 0.7$, half the variance is eliminated — equivalent to doubling your traffic.

Safety rule: Covariates must come from the *pre-experiment* period. Using post-treatment covariates can introduce bias and invalidate the analysis.

Step 4: Estimate the treatment effect

Compute the adjusted group means and the **relative lift**:

$$\hat{\tau}(t) = \hat{\mu}_1(t) - \hat{\mu}_0(t), \qquad \text{Relative lift}(t) = \frac{\hat{\tau}(t)}{\hat{\mu}_0(t)}$$

$hatmu_0(t)$ and $hatmu_1(t)$ are the adjusted group means at time t. Their difference $hatau(t)$ estimates the average treatment effect. Dividing by the control mean yields the relative lift.

Relative lift is preferred because it is **scale-invariant**— a 3% lift has the same interpretation regardless of the baseline level.

Step 5: Estimate the variance

$$\hat{\sigma}^2_{\hat{\tau}}(t) = \frac{s_0^2(t)}{n_0(t)} + \frac{s_1^2(t)}{n_1(t)}$$

Here $s_g^2(t)$ is the sample variance (spread) of the adjusted outcomes in group g, and $n_g(t)$is the number of users in that group so far. The variance is estimated from the data — not assumed known.

Step 6: Construct the sequential confidence interval

Instead of using the fixed critical value $z_{alpha/2} approx 1.96$, Eppo applies a time-varying multiplier derived from the confidence sequence framework:

$$\text{CI}(t) = \hat{\tau}(t) \ \pm \ \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{\sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}}_{\text{sequential multiplier}}$$

Unpacking the formula:

- $\hat{\tau}(t)$: the estimated treatment effect at time t (from Step 4).
- $\hat{\sigma}_{\hat{\tau}}(t)$: the estimated standard error (from Step 5).
- $n = n_0(t) + n_1(t)$: total observations so far.
- α: significance level (e.g. 0.05 for a 95% CI).

- ν: a tuning parameter set from the expected total sample size M: $\nu = M \cdot \hat{\sigma}^2$. It controls *which* sample size the CI is tightest for.
- The "sequential multiplier" replaces the constant 1.96. It starts higher (e.g. ~3.8 early on) and slowly decreases toward 1.96 — but never quite reaches it. This is the **price of peeking**.

This formula comes from the *confidence sequence* framework of Howard et al. (2021), specifically the Normal mixture boundary. The mathematical guarantee: the true treatment effect lies inside the CI at *all times simultaneously* with probability $\geq 1 - \alpha$.

Step 7: Decide

What you see	Colour	Action
CI entirely above 0	Green	Positive effect. Ship the feature.
CI entirely below 0	Dark grey	Negative effect. Revert.
CI crosses 0, experiment ongoing	Grey	Inconclusive. Keep collecting.
CI crosses 0, experiment ended	Grey	No significant effect. Decide pragmatically.

This decision is valid at any stopping time. No pre-specified analysis schedule is required. This resolves the peeking problem from Act 1.

Key Takeaway

The Eppo pipeline in one sentence: Randomise → collect outcomes Y and covariates X → regression-adjust (remove predictable noise) → estimate lift and variance → wrap in a sequential CI → decide when CI excludes zero.

What makes it work:

- **Framework:** Howard et al. (2021) confidence sequences.
- **Noise reduction:** Generalised CUPED via per-group regression.
- **Output:** Relative lift with a CI that is valid at any time.

The Hybrid Approach

The sequential CI is always wider than a fixed-horizon CI at any given sample size. For the **primary KPI**— the metric the experiment is designed to move — this width penalty reduces statistical power.

Apply sequential testing where early stopping adds the most value (safety guardrails), and fixed-horizon testing where statistical power matters most (the primary metric).

The hybrid approach partitions metrics into two categories:

- **Guardrail KPIs** (revenue, error rate, latency, etc.): monitored continuously with a **sequential CI**. If the CI excludes zero on the harmful side, abort the experiment immediately.
- **Primary KPI** (the metric the experiment targets): analysed with a **standard fixed-horizon CI** at the pre-planned end date. No peeking penalty. Full statistical power.

What you gain

	Full sequential	Hybrid
Primary KPI CI width	Wider (~10–40%)	Standard (no penalty)
Primary KPI power	Reduced	Full
Guardrail protection	Continuous	Continuous
Early stopping for success	Yes	No
Early stopping for harm	Yes	Yes
Complexity	Higher	Lower

Key Takeaway

The hybrid approach in one sentence: Monitor guardrail KPIs with a sequential CI for early abort; analyse the primary KPI with a standard CI at the planned end-date.

This gives you the safety of sequential testing where it matters most (preventing harm) without paying the power penalty on the metric you care about most (the primary KPI).

This is the recommended approach for most A/B testing programmes.

Act 3 — Alternative Methods

Three group sequential methods for controlling the family-wise error rate under interim analyses, implementable without a dedicated platform.

Suppose you plan to analyse your experiment at K pre-specified interim time points. Using the standard $\alpha = 0.05$ threshold at each analysis inflates the overall Type I error rate (Act 1).

The fix: **adjust the per-analysis significance level** so that the family-wise error rate remains at α. The three methods below differ in *how* they allocate the error budget across the K analyses.

- **Bonferroni:** equal allocation, α/K per analysis. Conservative because it ignores correlation across analyses.
- **Pocock:** constant critical value calibrated to the joint distribution of the test statistics across analyses. Tighter than Bonferroni.
- **O'Brien–Fleming:** front-loaded allocation. Very strict early, nearly standard at the final analysis.

Eppo's approach (Act 2) does not require pre-specifying K— it provides a continuously valid guarantee.

Number of planned analyses (K): 6

Method 1: Bonferroni

The simplest correction. Divide α by K. One line of code.

Method 1: Bonferroni Correction (Simplest)

The idea

Before the experiment starts, fix the number of planned analyses K (e.g. weekly for 4 weeks: $K = 4$). At each analysis, use a significance level of α/K instead of α.

The recipe

1. Choose K (number of planned peeks) and α (e.g. 0.05).
2. At each peek $k = 1, \ldots, K$, compute the standard CI using the adjusted significance level:

$$\text{CI}_k = \hat{\tau}(t_k) \ \pm \ \hat{\sigma}_{\hat{\tau}}(t_k) \cdot z_{\alpha/(2K)}$$

where $\hat{\tau}(t_k)$ is the estimated treatment effect at peek k, $\hat{\sigma}_{\hat{\tau}}(t_k)$ is its standard error, and $z_{\alpha/(2K)}$ is the critical value from a Normal distribution at the adjusted significance level.

3. If the CI does not cross 0: stop and declare significance.
4. If $k = K$ and the CI crosses 0: no significant effect.

Critical Values

K (peeks)	α/K	$z_{\alpha/(2K)}$	CI multiplier vs. 1.96
1	0.050	1.96	1.00× (no correction)
2	0.025	2.24	1.14×
4	0.0125	2.50	1.28×
8	0.00625	2.73	1.39×
12	0.00417	2.87	1.46×
20	0.0025	3.02	1.54×
52	0.00096	3.29	1.68× (weekly for a year)

With $K = 4$, replace $z = 1.96$ with $z = 2.50$. The CI is 28% wider at each analysis, but the family-wise error rate is controlled at $alpha$.

Implementation step by step

Given data arrays `control[]` and `treatment[]` at each scheduled peek:

1. **Compute group statistics:**

$$\bar{Y}_0 = \frac{1}{n_0}\sum_{i=1}^{n_0} Y_{0,i}, \qquad s_0^2 = \frac{1}{n_0 - 1}\sum_{i=1}^{n_0}(Y_{0,i} - \bar{Y}_0)^2$$

2. **Compute the estimated treatment effect and its standard error:**

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0, \qquad \mathrm{SE} = \sqrt{\frac{s_0^2}{n_0} + \frac{s_1^2}{n_1}}$$

3. **Look up the adjusted critical value:**

$$z^* = z_{\alpha/(2K)} = \Phi^{-1}\left(1 - \tfrac{\alpha}{2K}\right)$$

   where Φ^{-1} is the inverse of the standard Normal CDF (e.g. `scipy.stats.norm.ppf` in Python).

4. **Construct the CI and decide:**

$$\mathrm{CI} = \left[\hat{\tau} - z^* \cdot \mathrm{SE},\ \hat{\tau} + z^* \cdot \mathrm{SE}\right]$$

   If the CI does not contain 0, stop and declare significance. Otherwise, wait until the next scheduled peek.

Pros and cons

+ Dead simple. One line of code: change 1.96 to $z_{\alpha/(2K)}$.
+ Always valid. Works for any test statistic, any distribution.

– Conservative. The CI is wider than necessary because Bonferroni ignores the correlation between test statistics at successive peeks.
– Scales poorly. As K grows, the correction becomes increasingly harsh.

Method 2: Pocock (1977)

Constant critical boundary. Tighter than Bonferroni by exploiting correlation.

Method 2: Pocock Boundaries (1977)

The idea

Like Bonferroni, pre-specify K equally-spaced analysis times. But instead of Bonferroni's crude α/K split, use a constant critical value c_P that accounts for the correlation between test statistics at different peeks.

Because the test statistics $Z_1, Z_2, \ldots, Z_K$ at successive analyses share overlapping data, they are positively correlated. Pocock's method exploits this to derive a tighter threshold.

The recipe

1. Choose K and α.
2. Look up (or compute) the Pocock critical value $c_P(K, \alpha)$.
3. At each peek k, compute the z-statistic: $Z_k = \hat{\tau}(t_k)/\hat{\sigma}_{\hat{\tau}}(t_k)$.
4. If $|Z_k| > c_P$: stop and declare significance.

Pocock vs Bonferroni critical values

K	c_P (Pocock)	$z_{\alpha/(2K)}$ (Bonf.)	Improvement
2	2.18	2.24	3% tighter
4	2.36	2.50	6% tighter
8	2.51	2.73	8% tighter
12	2.60	2.87	9% tighter
20	2.69	3.02	11% tighter

Pocock gives a constant threshold that is a few percent lower than Bonferroni's. The key advantage: the threshold is the same at every peek, so it is almost as simple to use as Bonferroni.

How the boundary is computed

Under the null, the z-statistics at successive analyses follow a multivariate Normal with correlation:

$$\mathrm{Cor}(Z_j, Z_k) = \sqrt{\frac{\min(j,k)}{\max(j,k)}} \quad \text{for } 1 \le j, k \le K$$

The Pocock boundary c_P is the unique value satisfying:

$$\mathbb{P}\Big(\max_{1 \le k \le K} |Z_k| > c_P\Big) = \alpha$$

Once c_P is known, the decision rule at each peek is identical to Bonferroni, but with c_P replacing $z_{\alpha/(2K)}$.

Method 3: O'Brien–Fleming (1979)

Decreasing threshold. Nearly no penalty at the final analysis.

Method 3: O'Brien–Fleming Boundaries (1979)

The idea

Instead of a constant threshold, use a threshold that *decreases* over time. At the first peek, demand very strong evidence; at the last peek, demand nearly the same evidence as a classical test.

The recipe

At peek k (of K total), use the critical value:

$$c_k^{\mathrm{OBF}} \approx z_{\alpha/2} \cdot \sqrt{K/k}$$

where $z_{\alpha/2} \approx 1.96$ is the standard Normal critical value.

Critical values at each peek (K=4)

Peek k (of K=4)	c_k^{OBF}	c_P (Pocock)	$z_{\alpha/(2K)}$ (Bonf.)	$z_{\alpha/2}$ (class
1 (25% of data)	**4.05**	2.36	2.50	1.96
2 (50% of data)	**2.86**	2.36	2.50	1.96
3 (75% of data)	**2.34**	2.36	2.50	1.96
4 (100% of data)	**2.02**	2.36	2.50	1.96

O'Brien–Fleming is extremely conservative early (threshold 4.05 at the first analysis — early stopping is rare) but barely penalises the final analysis (2.02 vs. 1.96 — only 3% wider). This makes it the most efficient method: if the experiment runs to completion, the power loss from the interim analyses is negligible.

Implementation step by step

1. **Before the experiment:** Fix K and α. At each peek k, the fraction of total data collected is $t_k = k/K$ (assuming equal spacing).
2. **At peek k, compute the critical value:**

$$c_k^{\text{OBF}} = \frac{z_{\alpha/2}}{\sqrt{t_k}} = z_{\alpha/2} \cdot \sqrt{\frac{K}{k}}$$

3. **Compute the z-statistic** (same as Bonferroni and Pocock):

$$Z_k = \frac{\hat{\tau}(t_k)}{\text{SE}(t_k)}$$

4. **Decide:** If $|Z_k| > c_k^{\text{OBF}}$: reject. Otherwise: continue to peek $k+1$.
5. **Convert to a confidence interval** (optional but recommended):

$$\text{CI}_k = \hat{\tau}(t_k) \ \pm \ c_k^{\text{OBF}} \cdot \text{SE}(t_k)$$

Note: the CI width *changes* across peeks because c_k^{OBF} varies.

Practical tips

The exact O'Brien–Fleming boundaries (computed via multivariate Normal integration) differ slightly from the $z_{\alpha/2}/\sqrt{t_k}$ approximation. The approximation is accurate for $K \leq 10$ and is what most software implements. For larger K, use a library (e.g. `statsmodels` or the R package `gsDesign`).

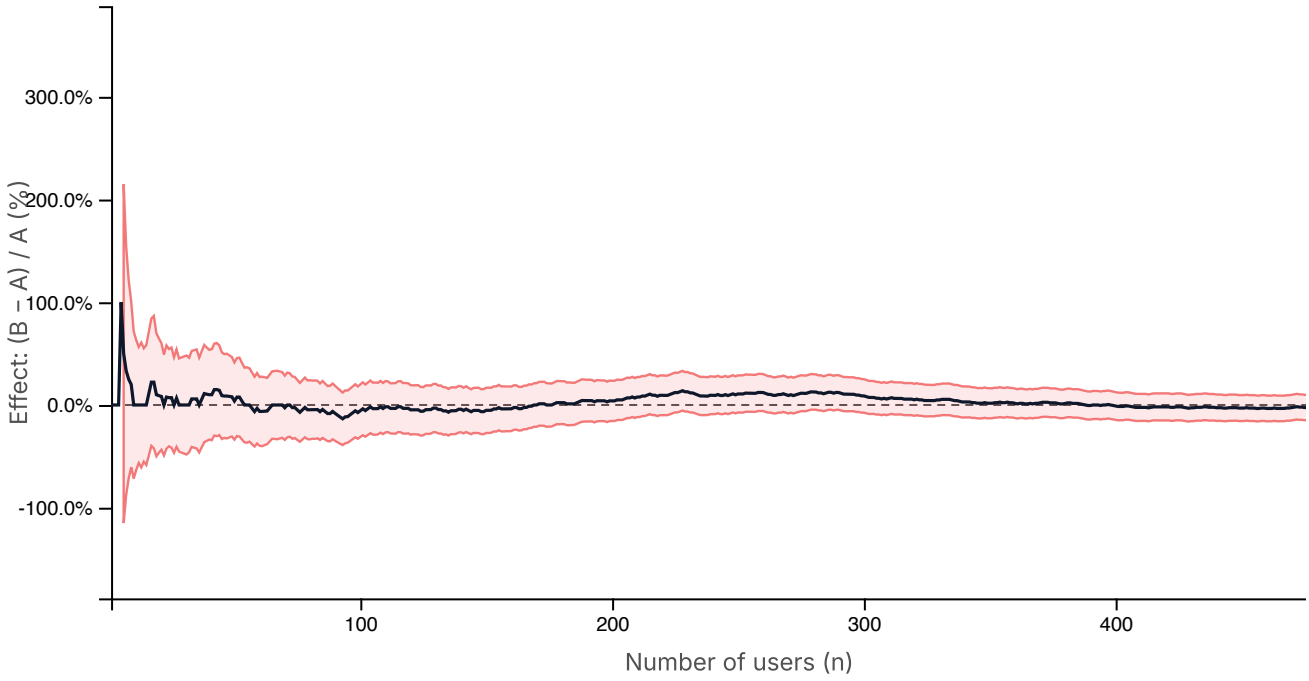
Simulation: Share Crossing Each Threshold

Effect: (B − A) / A, null hypothesis = 0%.

Effect size: +0.00

Number of users (n = 500) | α (0.05) | Effect size (+0.00)

Re-randomize | Standard 95% CI | Sample diff



SAMPLE EFFECT	CI HALF-WIDTH	DECISION AT N = 500
−3.50%	±12.06%	Inconclusive

Simulation takeaway. For each method, this shows the share of 500 simulated A/B tests (effect = 0) that crossed the respective threshold at any point. When effect = 0, this is an A/A test; otherwise, it shows the probability of crossing under the simulated effect.

Note: The Eppo/Howard sequential CI is very conservative in this setting, with a type I error rate often well below 1%. One of the group sequential alternatives, especially O'Brien–Fleming or Pocock, may be a better choice for most practical A/B tests, as they are not overly conservative and still control the error rate under peeking.

Comparison

	Bonferroni	Pocock	OBF	Eppo
Pre-specify K?	Yes	Yes	Yes	No
Threshold type	Constant	Constant	Decreasing	Continuous

	Bonferroni	Pocock	OBF	Eppo
CI at peek 1 (K=4)	$2.50 \times \hat{\sigma}$	$2.36 \times \hat{\sigma}$	$4.05 \times \hat{\sigma}$	$\sim 3.8 \times \hat{\sigma}$
CI at final analysis (K=4)	$2.50 \times \hat{\sigma}$	$2.36 \times \hat{\sigma}$	$2.02 \times \hat{\sigma}$	$\sim 2.3 \times \hat{\sigma}$
Valid between peeks?	No	No	No	Yes
Variance reduction?	Manual	Manual	Manual	Built-in
Implementation	1 line	Lookup table	Formula	Platform

Key Takeaway

Which method to use:

- **Minimal implementation effort:** Bonferroni. Replace 1.96 with $z_{\alpha/(2K)}$.
- **Best efficiency at the final analysis:**O'Brien–Fleming. The final-analysis critical value is within 3% of the unadjusted value, but requires pre-specifying K.
- **Continuous monitoring or built-in variance reduction:** Use Eppo or an equivalent platform.

Bottom line: even the simplest correction (Bonferroni) eliminates the bulk of the peeking-induced inflation. Any correction is vastly better than none.

Implementing the Hybrid Approach Without Eppo

Act 2 introduced the hybrid approach: sequential CI on guardrail KPIs for early abort, standard CI on the primary KPI at the planned end date. Below is how to implement it using any of the three correction methods above.

Step 1: Classify your metrics

Category	Examples	Testing method
Primary KPI	Conversion rate, sign-ups	Fixed-horizon ($z = 1.96$)
Guardrail KPIs	Revenue, error rate, latency, crash rate	Sequential (any of the 3 methods)

Step 2: Set guardrail significance levels

If you have G guardrail KPIs, set $\alpha_{\text{guardrail}} = \alpha/G$ for each one.

Example: 3 guardrails ($G = 3$), $\alpha = 0.05$, 4 weekly peeks ($K = 4$):

$$\alpha_{\text{per guardrail}} = \frac{0.05}{3} \approx 0.0167$$

Then apply Bonferroni within each guardrail:

$$z^* = \Phi^{-1}\left(1 - \tfrac{0.0167}{2\times 4}\right) = \Phi^{-1}(0.9979) \approx 2.86$$

Step 3: Run the experiment

At each scheduled peek:

1. For **each guardrail KPI**: compute the CI using your chosen sequential method.
2. If **any** guardrail CI is entirely on the harmful side of zero: **ABORT** the experiment.
3. Otherwise: continue.

At the **end** of the experiment:

1. For the **primary KPI**: compute a standard CI with the full α:

$$\text{CI}_{\text{primary}} = \hat{\tau} \ \pm \ z_{\alpha/2} \cdot \text{SE}$$

SHIP if the CI excludes zero. Otherwise: no significant effect.

Key Takeaway

The hybrid approach without Eppo:

- Apply Bonferroni across your G guardrail KPIs: $\alpha_g = \alpha/G$.
- Within each guardrail, use O'Brien–Fleming (best) or Bonferroni (simplest).
- Test the primary KPI once at the end with a standard CI — no correction needed.
- **This is the recommended approach for teams without a sequential testing platform.**

Summary

Act	Key message
1	Peeking inflates false positives. You need a test that is valid at every time you check.
2	Sequential confidence intervals (CIs) are valid at every moment. The hybrid approach uses sequential CIs for guardrails and classical CIs for the primary KPI.
3	DIY alternatives: Bonferroni, Pocock, O'Brien–Fleming. All widen the CI to control error when peeking.
4–6	Martingales and Ville's inequality: the math that makes peeking safe.
7–9	SPRT, mixture, and confidence sequences: the evolution of sequential testing and estimation.
10–13	Eppo's pipeline: regression adjustment, sequential CIs, and actionable decisions.

The core trade-off: Valid interim monitoring requires wider confidence intervals. The penalty depends on the method, but the benefit is freedom to peek and act at any time.

Method	Price of peeking
No correction (naive)	Invalid inference (20–30% false positives)
Bonferroni	CI ~28–68% wider (depending on K)
Pocock	CI ~20–37% wider
O'Brien–Fleming	CI ~3% wider at final analysis
Eppo (sequential CI)	CI ~10–40% wider, valid *continuously*
Hybrid (recommended)	No penalty on primary KPI; sequential penalty on guardrails only

Mathematical Reference

Notation

Symbol	Meaning	Where used
Y_i	Outcome for user i during the experiment	Act 2, Steps 2–3
X_i	Pre-experiment covariate for user i	Act 2, Step 3 (CUPED)
Y_i^*	Adjusted outcome after regression adjustment	Act 2, Step 3
$\hat{\tau}(t)$	Estimated treatment effect at time t	Act 2, Step 4
$\hat{\mu}_0(t),\ \hat{\mu}_1(t)$	Adjusted group means at time t	Act 2, Step 4
$\hat{\sigma}_{\hat{\tau}}(t)$	Estimated standard error of treatment effect	Act 2, Step 5
$s_g^2(t)$	Sample variance in group g at time t	Act 2, Step 5
$n_0(t),\ n_1(t)$	Number of users in control / treatment at time t	Throughout
n	Total observations: n_0(t) + n_1(t)	Act 2, Step 6
α	Significance level (typically 0.05)	Throughout
ν	Tuning parameter: \nu = M \cdot \hat{\sigma}^2	Act 2, Step 6 (sequential CI)
M	Expected total sample size	Act 2, Step 6
K	Number of pre-planned analysis times (peeks)	Act 3
k	Current peek number (1 \leq k \leq K)	Act 3
t_k	Information fraction at peek k: t_k = k/K	Act 3 (OBF)
Z_k	z-statistic at peek k: \hat{\tau}/\text{SE}	Act 3
$z_{\alpha/2}$	Standard Normal critical value (\approx 1.96)	Throughout

Symbol	Meaning	Where used
$z_{\alpha/(2K)}$	Bonferroni-adjusted critical value	Act 3 (Bonferroni)
c_P	Pocock constant critical value	Act 3 (Pocock)
c_k^{OBF}	O'Brien–Fleming critical value at peek k	Act 3 (OBF)
Φ^{-1}	Inverse standard Normal CDF (quantile function)	Act 3
ρ	Correlation between pre-experiment and experiment metrics	Act 2, Step 3
G	Number of guardrail KPIs	Act 3 (Hybrid)

Key Formulas

Standard Confidence Interval

$$\text{CI} = \hat{\tau} \ \pm \ z_{\alpha/2} \cdot \text{SE}$$

Valid only at one pre-specified time. The constant $z_{\alpha/2} \approx 1.96$ for a 95% CI.

Sequential Confidence Interval (Howard et al., 2021)

$$\text{CI}(t) = \hat{\tau}(t) \ \pm \ \hat{\sigma}_{\hat{\tau}}(t) \cdot \sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}$$

Valid at *all times simultaneously*. The multiplier starts high (~3.8) and decreases toward 1.96.

CUPED Variance Reduction

$$Y_i^* = Y_i - \hat{f}_g(X_i) \qquad \text{Var}(Y^*) = \text{Var}(Y)(1-\rho^2)$$

At $\rho = 0.7$, half the variance is eliminated.

Treatment Effect & Standard Error

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0 \qquad \mathrm{SE} = \sqrt{\frac{s_0^2}{n_0} + \frac{s_1^2}{n_1}}$$

Bonferroni Adjusted CI

$$\mathrm{CI}_k = \hat{\tau}(t_k) \ \pm \ z_{\alpha/(2K)} \cdot \mathrm{SE}(t_k)$$

O'Brien–Fleming Boundary

$$c_k^{\mathrm{OBF}} = \frac{z_{\alpha/2}}{\sqrt{t_k}} = z_{\alpha/2} \cdot \sqrt{\frac{K}{k}}$$

Pocock Boundary

$$\mathbb{P}\Big(\max_{1 \le k \le K} |Z_k| > c_P\Big) = \alpha$$

With correlation: $\mathrm{Cor}(Z_j, Z_k) = \sqrt{\min(j,k)/\max(j,k)}$

Detailed Explanations

Confidence Sequences

A confidence sequence is a sequence of confidence sets $(C_t)_{t \ge 1}$ such that the true parameter lies in C_t for *all* t simultaneously with high probability. Unlike a standard CI (valid at one time), a confidence sequence allows continuous monitoring. The price: wider intervals, especially early on.

Alpha Spending

The total false positive rate αcan be thought of as a “budget” that is spent across analyses. Bonferroni spends α/Kat each peek. O'Brien–Fleming spends almost nothing early and most at the end. The alpha-spending function $\alpha^*(t)$ formalises this: at information fraction t, cumulative type I error must not exceed $\alpha^*(t)$.

Information Fraction

The information fraction at peek k is $t_k = k/K$ for equally-spaced analyses, or more generally the ratio of current to planned total statistical information (proportional to sample size for equal variances). It ranges from 0 to 1 and determines where you are in the experiment.

FWER vs FDR

FWER (Family-Wise Error Rate): the probability of making *at least one* false positive across all tests. Bonferroni, Pocock, and OBF all control FWER. **FDR** (False Discovery Rate): the expected *proportion* of false positives among rejected hypotheses. FDR control is less strict and typically used when testing many hypotheses simultaneously (e.g. gene expression). For A/B testing with a small number of metrics, FWER control is standard.

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